

Bongjun Kim

• <https://www.bongjunkim.com>

EDUCATION	Ph.D. in Computer Science	June 2020
	Interactive Audio Lab Northwestern University, Evanston, IL, USA	
	M.S. in Culture Technology	Aug 2012
	Korea Advanced Institute of Science and Technology (KAIST), Korea	
	B.S. / M.S. in Industrial Engineering	Aug 2009
	Ajou University, Suwon, Korea	
	(Study Abroad, Lulea University of Technology, Lulea, Sweden)	Fall, 2006
RESEARCH INTERESTS	Machine Learning - deep learning, interactive machine learning, active learning	
	Audio Signal Processing - sound event recognition, audio/music information retrieval	
	Human Computer Interaction - interactive intelligent systems, creativity support tools	
PROFESSIONAL EXPERIENCE	Data Science Specialist Solventum, MN, USA	Apr 2024 – Present
	▪ Sound and speech recognition	
	Data Science Specialist 3M, MN, USA	Dec 2023 – Mar 2024
	Data Scientist	Aug 2020 – Nov 2023
	Health Information Systems	
	▪ Sound and speech recognition	
	Corporate Research Systems Lab AI Lab	
	▪ Sound event recognition	
	▪ Machine learning for manufacturing process	
	Research Assistant Northwestern University, IL, USA	Sep 2013 – Jun 2020
	Interactive Audio Lab. The Department of Computer Science	
	▪ Sound event detection with weakly-labeled data	
	▪ A human-in-the-loop interface for sound event annotation	
	▪ Audio search by vocal imitation	
	▪ Speeding learning of personalized audio equalization	
	Research Intern Bosch, CA, USA	Jun 2018 – Aug 2018
	The Research and Technology Center	
	▪ Deep learning for sound event recognition	
	Research Intern Gracenote, CA, USA	Jun 2017 – Sep 2017
	The Media Technology Lab.	
	▪ Audio compression identification	
	Research Assistant KAIST, Korea	Sep 2010 – Aug 2012
	Audio and Interactive Media Lab.	
	▪ Mobile media as a musical instrument	
	Research Engineer Doosan Infracore Co. Ltd, Korea	Jul 2009 – Jul 2010
	▪ Research process innovation	
TEACHING EXPERIENCE	Teaching Assistant Northwestern University	Spring, Fall 2019, Winter 2020
	CS-349 Machine Learning	

Teaching Assistant | Northwestern University
CS-352 Machine Perception of Music and Audio

Winter, 2019

Teaching Assistant | Northwestern University
EECS-349 Machine Learning

Fall quarters, 2014-2017

Teaching Assistant | KAIST, Korea
GCT-633 Audio and Multimedia Programming

Fall, 2012

**JOURNAL
PUBLICATIONS**

- Bryan Pardo, Mark Cartwright, Prem Seetharaman, and **Bongjun Kim**, “Learning to Build Natural Audio Production Interfaces,” *Arts*, Vol. 8, Issue 3, 2019
- **Bongjun Kim** and Bryan Pardo, “A Human-in-the-loop System for Sound Event Detection and Annotation ,” *ACM Transaction on Interactive Intelligent System (TiS)*, Vol. 8, Issue 2, Article 13, June 2018.
- **Bongjun Kim** and Kiejn Park, “Probabilistic Delay Model of Dynamic Message Frame in FlexRay Protocol,” *IEEE Transaction on Consumer Electronics*, Vol. 55, Issue 1, pp. 77-82, 2009.
- Bumjoo Park, Kiejn Park, and **Bongjun Kim**, “A Performance Isolation Mechanism Based on Fuzzy Technique for Web Server Loading Balancing,” *The Institute of Electronics, Information and Communication Engineers Transactions on Communications*, Vol.E92-B, No.4, 2009.

**REFEREED
CONFERENCE
AND WORKSHOP
PUBLICATIONS**

- Arindam Ghosh, Mark Fuhs, **Bongjun Kim**, Anurag Chowdhury and Monika Woszczyna, “ASR-Synchronized Speaker-Role Diarization,” *Interspeech*, 2026.
- Anurag Chowdhury, Abhinav Misra, Yinong Wang, **Bongjun Kim**, Mark Fuhs, and Monika Woszczyna, “Augmented State Space Speaker Clustering: Reformulating HMM Based Clustering To Improve Speaker Diarization,” *Odyssey*, 2026.
- **Bongjun Kim**, Arindam Ghosh, Mark Fuhs, Anurag Chowdhury, Deblin Bagchi, and Monika Woszczyna, “A Hybrid Approach to Combining Role Diarization with ASR for Professional Conversations,” *Interspeech*, 2025.
- Fatemeh Pishdadian, **Bongjun Kim**, Prem Seetharaman, and Bryan Pardo, “Classifying non-speech vocals: Deep vs Signal Processing Representations,” *the Detection and Classification of Acoustic Scenes and Events Workshop (DCASE)*, 2019.
- **Bongjun Kim** and Bryan Pardo, “Sound Event Detection Using Point-labeled Data,” *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2019.
- **Bongjun Kim** and Shabnam Ghaffarzadegan, “Self-supervised Attention Model for Weakly Labeled Audio Event Classification,” *European Signal Processing Conference (EUSIPCO)*, 2019.
- **Bongjun Kim** and Bryan Pardo, “Improving Content-based Audio Retrieval by Vocal Imitation Feedback,” *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2019.
- **Bongjun Kim**, Madhav Ghei, Bryan Pardo, and Zhiyao Duan, “Vocal Imitation Set: a dataset of vocally imitated sound events using the AudioSet ontology,” *the Detection and Classification of Acoustic Scenes and Events Workshop (DCASE)*, 2018.
- **Bongjun Kim** and Zafar Rafii, “Lossy Audio Compression Identification,” *European Signal Processing Conference (EUSIPCO)*, 2018.
- **Bongjun Kim**, “Leveraging User Input and Feedback for Interactive Sound Event Detection and Annotation,” Student Consortium, *ACM International Conference on Intelligent User Interfaces (IUI)* 2018.
- **Bongjun Kim** and Bryan Pardo, “I-SED: an Interactive Sound Event Detector,” *ACM International Conference on Intelligent User Interfaces (IUI)*, 2017
- **Bongjun Kim** and Bryan Pardo, “Interface Design for Interactive Sound Event Detection,” *Workshop on Awareness Interface and Interaction (AWARE) at the ACM International Conference on Intelligent User Interface (IUI)*, 2017

	▪ Bongjun Kim and Bryan Pardo, “Speeding Learning of Personalized Audio Equalization,” <i>International Conference on Machine Learning and Applications (ICMLA)</i>, Dec. 2014 ▪ Bongjun Kim and Bryan Pardo, “Adapting Collaborative Filtering to Personalized Audio Production,” <i>AAAI Conference on Human Computation and Crowdsourcing (HCOMP)</i>, 2014 ▪ Bongjun Kim and Woon Seung Yeo, “Probabilistic Prediction of Rhythmic Characteristics in Markov Chain-based Melodic Sequences,” <i>International Computer Music Conference (ICMC)</i> 2013 ▪ Seunghun Kim, Bongjun Kim, and Woon Seung Yeo, “IAMHear: A Tabletop Interface with Smart Mobile Devices using Acoustic Location,” <i>Conference on Human Factors in Computing Systems (CHI)</i> works in progress, 2013 ▪ Bongjun Kim and Woon Seung Yeo, “Interactive Mobile Music Performance with Digital Compass,” <i>the International Conference on New Interfaces for Musical Expression (NIME)</i>, 2012. ▪ Minkoo Kang, Kiejn Park, and Bongjun Kim, “A Scheduling Alogorithm for Reducing FlexRay Message Response Time using Empty Minislots in Dynamic Segment,” <i>Digest of Technical Papers, International Conference on Consumer Electronics (ICCE)</i>, 2010. ▪ Minkoo Kang, Kiejn Park, and Bongjun Kim, “A Static Message Scheduling Algorithm for Reducing FlexRay Network Utilization,” <i>IEEE International Symposium on Industrial Electronics</i>, 2009. ▪ Bongjun Kim and Kiejn Park, “Analysis of Frame Delay Probability in the FlexRay Dynamic Segment,” <i>The IEEE International Conference on Industrial Informatics</i>, 2008. ▪ Minkoo Kang, Kiejn Park, and Bongjun Kim, “PDO Packing Mechanism for Minimizing CANopen Network Utilization,” <i>The 34th Annual Conference of the IEEE Industrial Electronics Society</i>, 2008. ▪ Minkoo Kang, Kiejn Park, and Bongjun Kim, “Determining the Size of a Static Segment and Analyzing the Utilization of In-vehicle FlexRay Network,” <i>Third International Conference on Convergence and Hybrid Information Technology</i>, 2008.	
NON-REFEREED PUBLICATIONS	▪ Bongjun Kim, “Building Light-Weight Convolutional Neural Networks for Acoustic Scene Classification Using Audio Embeddings,” <i>the Detection and Classification of Acoustic Scenes and Events (DCASE) challenge</i>, 2021. ▪ Bongjun Kim, “Convolutional Neural networks with Transfer Learning for Urban Sound Tagging,” <i>the Detection and Classification of Acoustic Scenes and Events (DCASE) challenge</i>, 2019. ▪ Bongjun Kim, “Ensemble of Convolutional Neural Networks with Transfer Learning for Audio Classification,” <i>Making Sense of Sounds Data Challenge</i>, 2018.	
PATENTS	▪ M. Manner, L. Carlson, K. Kanukurthy, B. Kim, L. Kloc, G. Krueger, and G. Silsby, “Audio identification system for personal protective equipment,” WO2002/224062, Oct 2022. ▪ Zafar Rafii, Markus Cremer, and Bongjun Kim. “Methods, Apparatus and Articles of Manufacture to Identify Sources of Network Streaming Services,” US 11049507, June 2021. ▪ Zafar Rafii, Markus Cremer, and Bongjun Kim. “Methods, Apparatus and Articles of Manufacture to Identify Sources of Network Streaming Services,” US 10733998, August 2020. ▪ Zafar Rafii, Markus Cremer, and Bongjun Kim, “Methods and Apparatus to Perform Windowed Sliding Transforms,” US 10629213, April 2020.	
AWARDS/ HONORS	2 nd place in IEEE DCASE “Urban Sound Tagging” Challenge (10 teams)	2019
	IEEE Detection and Classification of Acoustic Scenes and Events Challenge	
	2 nd place in IEEE DCASE “Making Sense of Sound Data” Challenge (12 teams)	2018
	IEEE Detection and Classification of Acoustic Scenes and Events Challenge	
	WASPAA Travel Grant	2019
	IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)	

	The NSF Travel Grant	2019
	European Signal Processing Conference (EUSIPCO)	
	The SIGCHI Student Travel Grant	2019
	ACM International Conference on Intelligent User Interfaces (IUI)	
	The NSF Travel Grant	2017, 2018
	ACM International Conference on Intelligent User Interfaces (IUI)	
	Segal Design Cluster Fellowship	2015 – 2016
	Northwestern University, IL, USA	
	Outstanding Teaching Assistant Award	2012
	GCT633 Audio and Multimedia Programming	
	Korea Advanced Institute of Science and Technology (KAIST), Korea	
	Best Presentation and Paper Award	2008
	IEEE International Conference on Industrial Informatics	
INVITED TALKS / POSTERS	[Talk] “Interactive Intelligent Systems for Environmental Sound Event Search”	Mar 2024
	Kangwon National University, Korea	
	[Talk] “Interactive Intelligent Systems for Environmental Sound Event Search”	Nov 2022
	Applied AI Conference, St. Paul, MN, USA	
	[Talk] “Building AI models for sound recognition with less annotation effort”	Dec 2020
	AppliedAI meetup, St. Paul, MN, USA	
	[Poster] “Sound Event Detection using Point-labeled Data”	Oct 2019
	Speech and Audio in the Northeast (SANE), Columbia University, NYC, USA	
	[Talk] “A Human-in-the-loop System for Audio Retrieval”	Jun 2019
	Midwest Music and Audio Day, Indiana University, Bloomington, IN, USA	
	[Talk] “A Human-in-the-loop System for Sound Event Detection and Annotation”	Mar 2019
	ACM International Conference on Intelligent User Interfaces (IUI 2019), LA, USA	
	[Poster] “Vocal Imitation Set: a dataset of vocally imitated sound events”	Oct 2018
	Speech and Audio in the Northeast (SANE), Google, Cambridge, MA, USA	
	[Talk] “Interactive Intelligent user interfaces for Music and Audio”	Mar 2018
	HCI@KAIST Seminars, KAIST, Korea	
	[Talk] “Interactive Intelligent user interfaces for Music and Audio”	Mar 2018
	Music and Audio Computing Lab., Culture Technology, KAIST, Korea	
	[Talk] “Interactive Intelligent user interfaces for Music and Audio”	Mar 2018
	Ewha Arts & Science Institute, Ewha Womans University, Korea.	
	[Talk] “A Human-in-the-loop System for Sound Event Detection and Annotation”	Mar 2018
	Haii:Human-AI-Interaction, inc., Yonsei University, Korea	
	[Talk] “I-SED: an Interactive Sound Event Detector”	Jun 2017
	Midwest Music and Audio Day, Evanston, IL, USA	
	[Talk] “I-SED: an Interactive Sound Event Detector”	Feb 2017
	Simons Institute Workshop on Interactive Learning, Berkeley, CA, USA	
	[Talk] “I-SED: an Interactive Sound Event Detector”	Feb 2017
	Gracernote, Emeryville, CA, USA	

EXTERNAL SERVICES	Journal Reviewer	
	IEEE Signal Processing Letters	2020
	IEEE Signal Processing Magazine	2018
	Conference Reviewer	
	International Conference on Acoustics, Speech, and Signal Processing (ICASSP)	2015 – Present
	Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)	2021 – Present
	International Conference on New Interfaces for Musical Expression (NIME)	2013 – Present
	Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)	2015, 2021
	ACM User Interface Software and Technology Symposium (UIST)	2019
	ACM Multimedia Conference	2016
	Student Volunteer for Conferences	
	ACM International Conference on Intelligent User Interfaces (IUI), Cyprus	2017
	International Conference on Design Computing and Cognition, USA	2016
	Conference Organizing Staff	2013
	International Conference on New Interfaces for Musical Expression (NIME), Korea	
DATA SET	Vocal Imitation Set: vocal imitations of sounds from the AudioSet ontology https://doi.org/10.5281/zenodo.1340763	May 2018
	Fine-grained Vocal Imitation Set https://doi.org/10.5281/zenodo.3538534	Nov 2019
GUEST LECTURES	“Sound object labeling”	Feb 2019
	CS-352: Machine perception of Music and Audio, Northwestern University	
	“Collaborative Filtering”	Oct 2017
	EECS-349: Machine Learning, Northwestern University	
	“Collaborative Filtering”	Oct 2016
	EECS-349: Machine Learning, Northwestern University	
	“Collaborative Filtering”	Oct 2015
	EECS-349: Machine Learning, Northwestern University	
MEDIA	Podcast Guest	Mar 2021
	<i>Conversations on Applied AI - Stories from Experts in Artificial Intelligence</i> https://appliedai.buzzsprout.com/1101152/8222571	
RESEARCH MENTORING	Emma McDonnell Undergraduate at Northwestern Univ.	2019
	Project: Audio editing interfaces for the visually impaired	
	Brian Margolis M.S. at Northwestern Univ.	2018
	Project: Interactive audio annotation interfaces for multi-class labeling	
	Madhav Ghei Undergraduate at Northwestern Univ.	2018
	Project: A query-by-vocal imitation audio search system	
ART PERFORMANCE AND EXHIBITION	Mobile music performance: Where Are You Standing?	May 2012
	Conference on New Interfaces for Musical Expression (NIME), Ann Arbor, USA	
	Interactive sound art installation: Turning Into Sound	Jan 2012
	Daejoen Children Art Gallery, Daejeon, Korea	

Mobile music performance: ADC Project- Don't Imagine

Aug 2011

Arco Art Theater, Seoul, Korea

Mobile music performance: Where Are You Standing?

Jun 2011

Hyundai Card / Capital Inc., Seoul, Korea